The spectral wave model wim_simple (Wave Interaction Model, WIM)

wim_simple is a spectral surface wave model, coupled to an ocean model in an energetically and dynamically consistent way. Basis is the wave action spectral density equation for deep water surface gravity waves

$$\partial_t A + \dot{\boldsymbol{x}} \cdot \boldsymbol{\nabla_x} A + \dot{\boldsymbol{k}} \cdot \boldsymbol{\nabla_k} A \;\; = \;\; S = S_{in} + S_{out} + S_{nl} \tag{1}$$

with wave action density $A(\boldsymbol{k}, \boldsymbol{x}, t)$, the horizontal wavenumber vector $\boldsymbol{k} = (k_x, k_y)$, the dispersion relation $\Omega = \sigma + \boldsymbol{U} \cdot \boldsymbol{k}$ with intrinsic frequency $\sigma(\boldsymbol{k}) = \sqrt{g|\boldsymbol{k}|}$ and Doppler shift velocity $\boldsymbol{U}$, transport velocities $\dot{\boldsymbol{x}} = \boldsymbol{\nabla_k}\Omega$ and $\dot{\boldsymbol{k}} = -\boldsymbol{\nabla_x}\Omega$, action input by wave growth S_{in}, action dissipation by wave breaking S_{out}, and non-linear action transfer S_{nl} and operators $\boldsymbol{\nabla_x} = (\partial_x, \partial_y)$ and $\boldsymbol{\nabla_k} = (\partial_{k_x}, \partial_{k_y})$. Only deep water waves are considered, with $\boldsymbol{\nabla_x}\sigma = 0$ and $\partial_t \sigma = 0$, but the concept extends readily to waves at finite depth. Since the scattering integral S_{nl} conserves energy and momentum of the waves, it holds that $\int \sigma S_{nl} d\boldsymbol{k} = 0$ and $\int \boldsymbol{k} S_{nl} d\boldsymbol{k} = 0$.

The surface wave model predicts the Stokes velocity $\boldsymbol{u}_S$ and the wave stress $\boldsymbol{\tau}_o$ to force an ocean model. Considering the Lagrangian ocean velocity $\boldsymbol{u}_L = \boldsymbol{u} + \boldsymbol{u}_S$ and its kinetic energy, the consistent energy cycle becomes

$$\partial_t \int_{-\infty}^{0} |\boldsymbol{u}_L|^2/2 dz + \boldsymbol{\nabla_x} \cdot \int_{-\infty}^{0} \boldsymbol{u}_h^L \pi dz \;\; = \;\; C \tag{2}$$

$$\partial_t \int E d\boldsymbol{k} + \boldsymbol{\nabla_x} \cdot \int \left(\dot{\boldsymbol{x}} E + \dot{\boldsymbol{x}} A (\boldsymbol{U} \cdot \boldsymbol{k})\right) d\boldsymbol{k} \;\; = \;\; \int (\Omega S_{in} + \boldsymbol{U} \cdot \boldsymbol{k}\, S_{nl}) d\boldsymbol{k} + \int \sigma S_{out} d\boldsymbol{k} - C$$

with the wave energy spectral density $E = \sigma A$ and the exchange term

$$C \;\; = \;\; \int_{-\infty}^{0} \boldsymbol{u}_h^L \cdot \partial_t \boldsymbol{u}_S dz + \int_{-\infty}^{0} \boldsymbol{u}_h^L \cdot \partial_z \boldsymbol{\tau} dz = \int \boldsymbol{U} \cdot \boldsymbol{k} (\partial_t A - S_{out}) d\boldsymbol{k} \tag{3}$$

and the setting, following Vanneste and Young for the Doppler shift velocity $\boldsymbol{U}$,

$$\boldsymbol{U}(\boldsymbol{x}, \boldsymbol{k}, t) = \int_{-\infty}^{0} \boldsymbol{u}_h^L Q dz \;\; , \;\; \boldsymbol{u}_h^S(\boldsymbol{x}, z, t) = \int Q A \boldsymbol{k} d\boldsymbol{k} \;\; , \;\; Q(k, z) = 2k e^{2kz} \tag{4}$$

$$\partial_z \boldsymbol{\tau}(\boldsymbol{x}, z, t) = -\int Q \boldsymbol{k} S_{out} d\boldsymbol{k} \;\; , \;\; \boldsymbol{\tau}_o(\boldsymbol{x}, t) = \int_{-\infty}^{0} \partial_z \boldsymbol{\tau} dz = -\int \boldsymbol{k} S_{out} d\boldsymbol{k} \tag{5}$$

with $\int_{-\infty}^{0} Q\, dz = 1$ and the ocean momentum forcing $\boldsymbol{\tau}_o = -\int \boldsymbol{k} S_{out} d\boldsymbol{k}$, i.e. the stress on the ocean is the dissipation of wave momentum. Note that $\boldsymbol{U}$ contains both the Eulerian mean and the Stokes horizontal velocity components. Waves and ocean exchange energy by the effect of the Doppler shift and the ocean stress, both given by C; other source terms of the wave energy are the wave growth by the wind, $\int \Omega S_{in} d\boldsymbol{k}$, and the dissipation by

wave breaking, $\int \sigma S_{out} d\boldsymbol{k}$, which may be directed into a turbulent ocean energy reservoir (see below). The cycle closes exactly up to the term $\int \boldsymbol{U} \cdot \boldsymbol{k}\, S_{nl} d\boldsymbol{k}$, which represents the work done on the sheared mean flow by the vertical redistribution of Stokes momentum through the nonlinear transfer. Only the baroclinic part of $\boldsymbol{U}$ contributes to this term, since $\int \boldsymbol{k} S_{nl} d\boldsymbol{k} = 0$. The discrete interaction approximation used for S_{nl} evaluates resonant quadruplets based on the intrinsic frequency and thus enforces $\int \sigma S_{nl} d\boldsymbol{k} = 0$, such that with a background current this term appears as a spurious source; since it is small compared to the other sources, it is ignored.

Relation to WAM

The concept and discretisation of wim_simple follows closely the example of WAM found in WAM (cycle 7) at `https://github.com/myWAveModel/WAM`, except for simplified action sources. Instead of the canonical wavenumber coordinates $\boldsymbol{k}$ used above, frequency $\nu = \sigma/(2\pi)$ and wavenumber angle θ with $k_x = k \sin\theta$, $k_y = k\cos\theta$ and $k = |\boldsymbol{k}|$ are used as phase dimensions, covered by an equidistant grid in θ and a logarithmic grid in ν as in WAM. Different to WAM, however, ν is here the *intrinsic* frequency, while in WAM the extrinsic frequency is used; the intrinsic frequency is better suited to represent wave–mean flow interaction, following the example of WAVEWATCH III. Instead of the spectral energy density $E(\boldsymbol{k}) = \sigma A(\boldsymbol{k})$, the model carries the WAM-like spectral density $E(\theta, \nu)$, related by $\int_{-\infty}^{0} |\boldsymbol{u}_L|^2/2 dz = \int E(\boldsymbol{k}) d\boldsymbol{k} = g \int E(\theta,\nu) d\theta d\nu$. Units of $E(\boldsymbol{k}) d\boldsymbol{k}$ are m^3/s^2, thus $E(\theta,\nu) d\theta d\nu$ has units m^2 (the surface elevation variance) and $E(\theta,\nu)$ has units $m^2 s$, while $E(\boldsymbol{k})$ has units m^5/s^2; the factor g shows up accordingly in the definitions of $\boldsymbol{u}_S$ and $\boldsymbol{\tau}$ below.

The model equation formulated for $A(\theta,\nu)$ reads

$$\partial_t A + \boldsymbol{\nabla}_{\boldsymbol{x}} \cdot (\dot{\boldsymbol{x}} A) + \partial_\theta (\dot\theta A) + \partial_\nu (\dot\nu A) \;\; = \;\; S_{in} + S_{out} + S_{nl} \tag{6}$$

We need expressions for group velocity $\dot{\boldsymbol{x}}$ and the drift velocities $\dot\theta$ and $\dot\nu$.

Group velocity

Since $\boldsymbol{U} = \boldsymbol{U}(k)$, the group velocity is given by

$$\dot{\boldsymbol{x}} = (c/2)\boldsymbol{s} + \boldsymbol{U} + \boldsymbol{k} \cdot \boldsymbol{\nabla}_{\boldsymbol{k}} \boldsymbol{U} \tag{7}$$

with the phase velocity $c = \sigma/k = \sqrt{g}/\sqrt{k} = g/\sigma$ and the wave direction vector $\boldsymbol{s} = (\sin\theta, \cos\theta)$. Since the wavenumber gradient of the Doppler velocity $\boldsymbol{U}$ is purely radial (it points along $\boldsymbol{k}$),

$$\boldsymbol{k} \cdot \boldsymbol{\nabla}_{\boldsymbol{k}} \boldsymbol{U} = \begin{pmatrix} (k_x \partial_k U + k_y \partial_k V)\, k_x/k \\ (k_x \partial_k U + k_y \partial_k V)\, k_y/k \end{pmatrix} = (\boldsymbol{s} \cdot \partial_k \boldsymbol{U})\; \boldsymbol{k} = k\,(\sin\theta\, \partial_k U + \cos\theta\, \partial_k V)\; \boldsymbol{s}, \tag{8}$$

with $\partial_k \boldsymbol{U} = (\partial_k U, \partial_k V)$. The full group velocity reads

$$\dot{\boldsymbol{x}} = (c/2)\, \boldsymbol{s} + \boldsymbol{U} + k\,(\sin\theta\, \partial_k U + \cos\theta\, \partial_k V)\; \boldsymbol{s}. \tag{9}$$

With $k = (2\pi\nu)^2/g$ one has $k\,\partial_k U_i = k\partial_\nu U_i \partial\nu/\partial k = \frac{\nu}{2}\,\partial_\nu U_i$ so the radial correction can be written with a single scalar factor

$$R = \frac{\nu}{2}\,(\boldsymbol{s}\cdot\partial_\nu \boldsymbol{U})\,, \qquad \dot{\boldsymbol{x}} = (c/2)\boldsymbol{s} + \boldsymbol{U} + R\boldsymbol{s} \tag{10}$$

i.e. the same directional projection $R\,\boldsymbol{s}$ enters both components. This reduces to the pure group velocity $\dot{\boldsymbol{x}} = (c/2)\boldsymbol{s} + \boldsymbol{U}$ whenever the current has no frequency dependence ($\partial_\nu \boldsymbol{U} = 0$), e.g. for a barotropic current.

Other drift velocities

Since $\boldsymbol{U} = \boldsymbol{U}(k)$, the material derivative of $\boldsymbol{U}$ along a ray contains, besides $\partial_t \boldsymbol{U} + \dot{\boldsymbol{x}}\cdot\boldsymbol{\nabla}_{\boldsymbol{x}}\boldsymbol{U}$, also the contribution $\dot{\boldsymbol{k}}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}\boldsymbol{U}$, so $\dot\sigma$ becomes with $\dot{\boldsymbol{k}} = -(\boldsymbol{\nabla}\boldsymbol{U})\cdot\boldsymbol{k}$

$$\dot\sigma \;=\; \partial_t\Omega - \dot{\boldsymbol{U}}\cdot\boldsymbol{k} - \boldsymbol{U}\cdot\dot{\boldsymbol{k}} = (\dot{\boldsymbol{x}} - \boldsymbol{U})\cdot\dot{\boldsymbol{k}} - (\dot{\boldsymbol{k}}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}\boldsymbol{U})\cdot\boldsymbol{k} \tag{11}$$

With $\boldsymbol{\nabla}_{\boldsymbol{k}} U_i = (\partial_k U_i)\,\boldsymbol{k}/k$ and $k\,\partial_k U_i = (\nu/2)\partial_\nu U_i$ one finds $(\dot{\boldsymbol{k}}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}\boldsymbol{U})\cdot\boldsymbol{k} = (\dot{\boldsymbol{k}}\cdot\boldsymbol{s})\,k\,(\boldsymbol{s}\cdot\partial_k\boldsymbol{U}) = R\,(\boldsymbol{s}\cdot\dot{\boldsymbol{k}})$ with R from Eq. (10). Since also $\dot{\boldsymbol{x}} - \boldsymbol{U} = (c/2 + R)\,\boldsymbol{s}$ by Eq. (10), we obtain

$$\dot\sigma = 2\pi\dot\nu = ((c/2 + R)\,\boldsymbol{s})\cdot\dot{\boldsymbol{k}} - R\,(\boldsymbol{s}\cdot\dot{\boldsymbol{k}}) = (c/2)\,\boldsymbol{s}\cdot\dot{\boldsymbol{k}} \tag{12}$$

Finally,

$$\dot\theta = (\cos\theta\dot{k}_x - \sin\theta\dot{k}_y)k^{-1} \tag{13}$$

results using $\tan\theta = k_x/k_y$ and $\dot{k}_x = -k_x\partial_x U - k_y\partial_x V$ and $\dot{k}_y = -k_x\partial_y U - k_y\partial_y V$.

High frequency cut-off

Since the growth rates γ_{in} and γ_{out} increase strongly with frequency, which would require a very small time step at large ν, the wave model is only calculated up to a cut-off frequency ν_c, and for the higher frequencies $\nu > \nu_c$ a Phillips spectrum with a ν^{-5} tail of the wave spectrum is applied, the so-called high-frequency tail. It has the form $E_{hi}(\nu) = E(\nu_c,\theta)\nu_c^5\nu^{-5}$. The cut-off frequency ν_c is calculated as in WAMDI (1988)

$$\nu_c = \max(2.5\bar\nu, 3\nu_{pm}) \tag{14}$$

with the mean frequency $2\pi\bar\nu = \int E d\theta d\nu / \int (E/\sigma) d\theta d\nu$, the "wind sea" frequency $2\pi\nu_{pm} = g/(28u_\star)$, and the friction velocity $u_\star$. For $\nu > \nu_c$ it is assumed that Eq. (6) becomes local and steady, such that the source terms balance locally in space and time and in ν and θ, i.e. $S|_{\nu>\nu_c} = 0$. The per-column prognostic cut-off and the imposed ν^{-5} tail are activated by the attribute `mask_energy` (default off); the resulting prognostic/diagnostic bin mask is also the basis of the equilibrium tail-flux booking of `tail_flux = 'input'` described below, which therefore requires `mask_energy` to be active.

Source terms

The source terms S_{in} and S_{out} are parameterized by growth rates, $S_{in} = \gamma_{in} A$ and $S_{out} = \gamma_{out} A$, for which the simplest forms following WAMDI (1988) are used

$$\gamma_{out} = -\beta_o g^{-4} \bar{E}^2 \bar{\sigma}^7 \ \sigma^2 \quad , \quad \gamma_{in} = \sigma \frac{\rho_a}{\rho_o} \times 0.25 \max\left(0, 28(u^*/c)\cos(\theta - \theta*) - 1\right) \tag{15}$$

with mean energy $\bar{E} = \int E d\theta d\nu$, mean frequency $\bar{\sigma} = \bar{E} / \int (E/\sigma) d\theta d\nu$, air,water density ρ_a, ρ_o, atmospheric friction velocity magnitude $u^\star$, with direction θ^*, deep water phase velocity $c = g/\sigma$, and the parameter $\beta_o = 2.56$. Other parameterisations are implemented as options. The mean frequency is clipped to the resolved spectral range,

$$\bar{\sigma} \leftarrow \min(\max(\bar{\sigma},\, \sigma_0),\, \sigma_{N_\nu - 1}) \tag{16}$$

The non-linear source term S_{nl} is the discrete interaction approximation (DIA) of Hasselmann et al. (1985), taken from WAM (cycle 7). A fortran version directly from the original WAM legacy code, and two python versions (one with jax/jit acceleration, one pure numpy) are available. The attribute "snonlin_backend" determines which one to use ('fortran'/'jax'/'numpy'), default is 'fortran'; all versions give identical results.

Stokes velocity

Stokes velocity is

$$\boldsymbol{u}_S(z) = \int\int 2(k/\sigma) e^{2kz} \boldsymbol{k} E(\boldsymbol{k}) d\boldsymbol{k} = (2/g) \int_0^\infty \sigma^3 e^{2kz} \int_0^{2\pi} \boldsymbol{s} E d\theta d\nu \tag{17}$$

with direction vector $\boldsymbol{s} = (\sin\theta, \cos\theta)$. The frequency integral is calculated over the prognostic range of ν, and the contribution by the high frequency tail is added. The latter part is given by

$$\boldsymbol{u}_S^{hi} = (2/g) \int_{\nu_c}^\infty \int_0^{2\pi} \sigma^3 e^{2z\sigma^2/g} \boldsymbol{s} E(\nu_c, \theta) \nu_c^5 \nu^{-5} d\theta d\nu \approx (2/g) \nu_c \int_0^{2\pi} \sigma_c^3 \boldsymbol{s} E(\nu_c, \theta) d\theta \tag{18}$$

assuming that the uppermost model level thickness consumes the factor $e^{2z\sigma^2/g}$ and with $\sigma_c = 2\pi\nu_c$. Below the uppermost vertical level, $\boldsymbol{u}_S^{hi} = 0$ in this approximation. For small effective vertical decay scale of the tail contribution or large vertical levl thickness, this approximation gets worse. The frequency integral in Eq. (18) can in fact be carried out exactly, retaining the full depth dependence of the exponential. With $\sigma_c = 2\pi\nu_c$ the result is

$$u_S^{hi}(z) = \frac{2}{g} \nu_c \int_0^{2\pi} \mathbf{s}\, E(\nu_c, \theta)\, \sigma_c^3 \left[e^{2z\sigma_c^2/g} - \sigma_c \sqrt{-\frac{2\pi z}{g}}\ \mathrm{erfc}\left(\sigma_c \sqrt{-\frac{2z}{g}} \right) \right] d\theta, \tag{19}$$

valid for $z \le 0$. The approximation Eq. (18) is recovered in the near-surface limit $\sigma_c^2 |z|/g \ll 1$, where the erfc term vanishes and $e^{2z\sigma_c^2/g} \to 1$, leaving the bracket equal to σ_c^3 and hence the prefactor $\frac{2}{g}\nu_c \sigma_c^3$ of Eq. (18). Note that WAM (cycle 7) uses the approximated version; the difference grows with the ratio of the uppermost level thickness to the vertical decay scale of the tail.

Wave stress

The stress force in the ocean is given, with the dissipation energy source $\dot{E}_{out} = \gamma_{out} E$ in (θ, ν) coordinates, by

$$\partial_z \boldsymbol{\tau} = -(2/g) \int_0^\infty e^{2kz} \sigma^3 \int_0^{2\pi} \boldsymbol{s} \dot{E}_{out} d\theta d\nu \tag{20}$$

The stress itself is

$$\boldsymbol{\tau}_o = -\int_0^\infty \sigma \int_0^{2\pi} \boldsymbol{s} \dot{E}_{out} d\theta d\nu \tag{21}$$

The contribution by the high-frequency tail assuming the WAMDI dissipation scale is

$$\partial_z \boldsymbol{\tau}^{hi} = 2\nu_c^5 \beta_o g^{-5} \bar{E}^2 \bar{\sigma}^7 (2\pi)^5 \int_{\nu_c}^\infty e^{(8\pi^2/g) z \nu^2} d\nu \int_0^{2\pi} \boldsymbol{s} E(\nu_c, \theta) d\theta \tag{22}$$

with $E_{hi}(\nu) = E(\nu_c, \theta) \nu_c^5 \nu^{-5}$ and $\gamma_{out} = -\beta_o g^{-4} \bar{E}^2 \bar{\sigma}^7 \sigma^2$ and the finite integral

$$\int_a^\infty e^{-bx^2} dx = \frac{\sqrt{\pi} - \sqrt{\pi} \mathrm{erf}\left(a\sqrt{b}\right)}{2\sqrt{b}} \tag{23}$$

For the stress we find for the high-frequency tail

$$\boldsymbol{\tau}_o^{hi} = (2\pi)^3 \nu_c^4 \beta_o g^{-4} \bar{E}^2 \bar{\sigma}^7 \int_0^{2\pi} \boldsymbol{s} E(\nu_c, \theta) d\theta \tag{24}$$

An alternative formulation of the tail stress follows from the equilibrium assumption that underlies the imposed ν^{-5} shape itself: for $\nu > \nu_c$ the sources balance, $\dot{E}_{in} + \dot{E}_{nl} + \dot{E}_{out} = 0$, so the flux to the ocean, $-\dot{E}_{out}$, can be evaluated as $\dot{E}_{in} + \dot{E}_{nl}$ instead of extending the WAMDI dissipation scale (which is calibrated on the resolved spectrum and underestimates the dissipation that maintains the equilibrium range). The input part is

$$\boldsymbol{\tau}_o^{hi,in} = \int_{\nu_c}^\infty \sigma \, \gamma_{in} \, \nu_c^5 \nu^{-5} \, d\nu \int_0^{2\pi} \boldsymbol{s} \, E(\nu_c, \theta) \, d\theta \tag{25}$$

with γ_{in} from Eq. (15). Writing $28(u^\star/c)\cos(\theta - \theta^*) = a\nu$ with $a(\theta) = 28u^\star 2\pi \cos(\theta - \theta^*)/g$, the input is active only above $\nu_1 = \max(\nu_c, 1/a)$, and the frequency integral is elementary, $\int_{\nu_1}^\infty (a\nu^{-2} - \nu^{-3}) \, d\nu = a/\nu_1 - 1/(2\nu_1^2)$, giving per direction

$$\boldsymbol{\tau}_o^{hi,in} = \frac{\rho_a}{\rho_o} \frac{(2\pi)^2 \nu_c^3}{4} \int_0^{2\pi} \boldsymbol{s} \, E(\nu_c, \theta) \left\{ \begin{array}{ll} a\nu_c - \frac{1}{2}, & a\nu_c \geq 1 \\ \frac{1}{2}(a\nu_c)^2, & 0 < a\nu_c < 1 \end{array} \right\} d\theta \tag{26}$$

(continuous at $a\nu_c = 1$; zero for $\cos(\theta - \theta^*) \leq 0$). The integrand decays like ν^{-2}, i.e. the integral converges exactly as benignly as the WAMDI form in Eq. (24); note that the

corresponding integral with the dissipation of Janssen (1994; see also Komen et al. 1994, and the ECMWF IFS documentation, Part VII) diverges strongly because of its $(k/\bar{k})^4$ term – its $(k/\bar{k})^2$ term behaves like the WAMDI form – which is the reason the WAMDI scale is used in Eq. (24). The nonlinear part uses the conservation of the exact transfer over the full spectrum, $\int_0^\infty \sigma \int \boldsymbol{s}\, \dot{E}_{nl}\, d\theta\, d\nu = 0$, so that

$$\boldsymbol{\tau}_o^{hi,nl} = \int_{\nu_c}^{\infty} \sigma \int_0^{2\pi} \boldsymbol{s}\, \dot{E}_{nl}\, d\theta\, d\nu = -\int_0^{\nu_c} \sigma \int_0^{2\pi} \boldsymbol{s}\, \dot{E}_{nl}\, d\theta\, d\nu \tag{27}$$

i.e. the cascade flux into the tail equals minus the resolved-range integral of the computed nonlinear transfer (the "cascade credit"). For completeness, the resulting total stress on the ocean in this formulation is

$$\boldsymbol{\tau}_o = -\int_0^{\nu_c} \sigma \int_0^{2\pi} \boldsymbol{s} \left(\dot{E}_{out} + \dot{E}_{nl}\right) d\theta\, d\nu \;+\; \boldsymbol{\tau}_o^{hi,in} \tag{28}$$

where the first term is the resolved dissipation of Eq. (21) restricted to the prognostic range with the cascade credit Eq. (27) folded in, and the second term is the analytic tail input Eq. (26).

The depth-distributed version (the "stress work" of Eq. (20)) follows with the same replacement of the tail rate. With $-\dot{E}_{out} = \gamma_{in} E(\nu_c, \theta) \nu_c^5 \nu^{-5}$ above ν_c and $\sigma^3 \gamma_{in} = \frac{\rho_a}{\rho_o} \frac{(2\pi)^4}{4} \nu^4 (a\nu - 1)$,

$$\partial_z \boldsymbol{\tau}^{hi,in} = \frac{2}{g} \frac{\rho_a}{\rho_o} \frac{(2\pi)^4}{4} \nu_c^5 \int_0^{2\pi} \boldsymbol{s}\, E(\nu_c, \theta) \int_{\nu_1}^{\infty} e^{(8\pi^2/g) z\, \nu^2} \left(a - \nu^{-1}\right) d\nu\, d\theta \tag{29}$$

with $\nu_1 = \max(\nu_c, 1/a)$ as before. The a–term is the finite integral Eq. (23); the second term requires the companion

$$\int_{a_0}^{\infty} x^{-1} e^{-bx^2} dx = \frac{1}{2} E_1(a_0^2\, b) \tag{30}$$

with the exponential integral E_1. The vertical integral of Eq. (29) recovers Eq. (26) exactly. The cascade contribution Eq. (27) is deposited at the surface: it is dissipated by tail waves with $\nu \geq \nu_c$, whose deposition scales $g/2\sigma^2$ are centimetres, far below any ocean grid spacing. Both formulations are implemented, selected by the switch `tail_flux` $\in$ {`'wamdi'`, `'input'`}; the discrete treatment, including the diagnostic bins inside the grid, is described below.

Discretisation of the wave–ocean coupling and the energy exchange

All three coupling quantities of Eq. (4) – the Stokes velocity $\boldsymbol{u}_S$, the stress force $\partial_z \boldsymbol{\tau}$ and the Doppler velocity $\boldsymbol{U}$ – involve the same vertical kernel $Q(k, z) = 2k e^{2kz}$ with $\int_{-\infty}^{0} Q\, dz = 1$: $\boldsymbol{u}_S$ and $\partial_z \boldsymbol{\tau}$ assign spectral quantities to a depth profile, while $\boldsymbol{U}$ projects the Lagrangian velocity back onto the spectral grid. The exchange term C of Eq. (3) closes because $\boldsymbol{U}$ is the

adjoint of the profile assignment, i.e. because the same Q appears on both sides of the depth integral. The discretisation preserves this structure: one common discrete kernel is used in all three methods of `./pyOM3/wim/interface.py` (`calc_stokes`, `calc_stress_work`, `calc_doppler_vel`), given by the cell integral of Q over each ocean cell of thickness Δz with centres z_k and faces $z_{k\pm1/2} = z_k \pm \Delta z/2$ (uppermost face at $z = 0$),

$$\bar{Q}_k \Delta z = \int_{z_{k-1/2}}^{z_{k+1/2}} Q \, dz = e^{2kz_{k+1/2}} - e^{2kz_{k-1/2}} \tag{31}$$

in place of the point value $Q(z_k)\Delta z$. The cell integral is essential: the decay scale $g/(2\sigma^2)$ of the kernel is of the order of centimetres for the high frequencies of the prognostic range and for the tail, far below any ocean level thickness, so that point sampling loses a large part of the coupling amplitude (about half of the stress force for $\Delta z = 1\,m$ and a developing wind sea). The weights Eq. (31) telescope over the column, $\sum_k \bar{Q}_k \Delta z = 1 - e^{-2kD}$ for column depth D, with three consequences that hold exactly for any Δz: the column-integrated stress force equals the surface stress $\boldsymbol{\tau}_o$ of Eq. (21), the discrete Stokes transport $\sum_k \boldsymbol{u}_S(z_k)\Delta z$ equals its analytic value, and a depth-independent (barotropic) current is projected onto the Doppler velocity without attenuation, $\boldsymbol{U} = \boldsymbol{u}_L(1 - e^{-2kD})$. The frequency derivative $\partial_\nu \boldsymbol{U}$ entering the radial refraction term Eq. (10) is taken as the analytic derivative of the discrete kernel, $\partial_\nu(\bar{Q}_k \Delta z) = (2k/\nu)\, 2(z_{k+1/2} e^{2kz_{k+1/2}} - z_{k-1/2} e^{2kz_{k-1/2}})$, so that refraction sees the same Doppler projection as the advection.

For the high-frequency tails the same principle is applied to the exact tail profiles. The tail of the stress force Eq. (22) contains the frequency integral $W(t) = \int_{\nu_c}^{\infty} e^{-bt\nu^2} d\nu$ with $t = -z \geq 0$ and $b = 8\pi^2/g$, evaluated with Eq. (23); its antiderivative is

$$H(t) = \int_0^t W(t')dt' = \frac{\sqrt{\pi}}{b} \left[\sqrt{bt}\, \mathrm{erfc}\left(\nu_c \sqrt{bt}\right) + \frac{1 - e^{-\nu_c^2 bt}}{\nu_c \sqrt{\pi}} \right] \tag{32}$$

and each cell receives $\left(H(-z_{k-1/2}) - H(-z_{k+1/2})\right)/\Delta z$ in place of the point value. Since $H(\infty) = 1/(b\nu_c)$, the column sum of the tail force equals its exact depth integral $\boldsymbol{\tau}_o^{hi}$ of Eq. (24), using $2(2\pi)^5 \nu_c^5 g^{-5} H(\infty) = (2\pi)^3 \nu_c^4 g^{-4}$; the prefactor $(2\pi)^5$ in Eq. (22) is essential for this consistency. The Stokes tail Eq. (19) is treated the same way: with $a = 2\sigma_c^2/g$ its depth profile $B(t) = \nu_c[e^{-at} - \sqrt{\pi at}\, \mathrm{erfc}(\sqrt{at})]$ has the antiderivative ($S = \sqrt{t}$)

$$I(t) = \nu_c \left[\frac{1 - e^{-at}}{a} - 2\sqrt{\pi a} \left(\frac{S^3}{3} \mathrm{erfc}(\sqrt{a} S) + \frac{1 - (aS^2 + 1)e^{-aS^2}}{3\sqrt{\pi}\, a^{3/2}} \right) \right] \tag{33}$$

with $I(\infty) = \nu_c/(3a)$, so that the discrete tail transport equals the exact value $(2\pi/3)\nu_c^2 \int \boldsymbol{s} E(\nu_c, \theta) d\theta$. In tests the column sums reproduce $\boldsymbol{\tau}_o$ and the Stokes transport to machine precision; for a shallow column below a developed wind sea, however, a substantial part of the retained fraction $1 - e^{-2kD}$ of the wave momentum flux can be deposited below the column bottom. For a closed momentum budget the attribute `closed_bottom` extends the integration of the bottom cell to $z \to -\infty$, in the shared kernel ($e^{2kz_{bot}} \to 0$ in Eq. (31)) as well as in both

tail antiderivatives ($H, I \to H(\infty), I(\infty)$), so that $\sum_k \bar{Q}_k \Delta z = 1$ exactly: the bottom cell absorbs the deep part of all coupling profiles, no wave momentum is lost, and the barotropic Doppler projection becomes exact, $\boldsymbol{U} = \boldsymbol{u}_L$, at all frequencies (with the open bottom the low-frequency bins of a shallow column see an almost vanishing Doppler current). Since the kernel remains shared, the adjoint identity Eq. (34) below is unaffected. The closure amounts to assuming $\boldsymbol{u}_L$ constant below the column floor and is the recommended setting for shallow test columns; for deep columns ($2kD \gg 1$) it makes no difference.

The energy exchanged between ocean and waves is controlled by the same kernel. On the ocean side, with the adjoint pair Eq. (31) the discrete identity

$$\sum_k \boldsymbol{u}_h^L(z_k) \cdot \partial_t \boldsymbol{u}_S(z_k)\, \Delta z = \int \boldsymbol{U} \cdot \boldsymbol{k}\; \partial_t A \; d\boldsymbol{k} \tag{34}$$

holds exactly (to machine precision) for any spectral tendency $\partial_t A$ and any Δz, over the prognostic frequency range (the tails are diagnostic and remain outside this closure). The ocean therefore receives, per coupling step, exactly the energy $\int \boldsymbol{U} \cdot \boldsymbol{k}\, \partial_t A\, d\boldsymbol{k}$ corresponding to the actual discrete evolution of the wave field, and the handoff itself introduces no discretisation error.

On the wave side, the counterpart of this energy must be extracted from (or added to) the intrinsic wave energy $\int \sigma A\, d\boldsymbol{k}$ by the propagation operator. For steady $\boldsymbol{U}$ the rays conserve $\Omega = \sigma + \boldsymbol{U} \cdot \boldsymbol{k}$, so the continuous propagation operator satisfies $\int \Omega\, \partial_t A|_{prop}\, d\boldsymbol{k} = 0$ and the intrinsic energy change $\int \sigma\, \partial_t A$ balances the ocean gain $-\int \boldsymbol{U} \cdot \boldsymbol{k}\, \partial_t A$ identically. Discretely, the flux-form advection conserves the action $\int A\, d\boldsymbol{k}$ to machine precision, but not $\int \Omega A\, d\boldsymbol{k}$: the nonlinear Superbee limiter leaves a residual (the frequency advection velocity $\dot{\nu}$ is already located on the u-centred frequency faces, so face staggering is not the cause). In tests with a wave packet refracted across a jet (moderate Doppler shift, $\max|\boldsymbol{U} \cdot \boldsymbol{k}|/\sigma \approx 0.3$) this Ω-conservation defect amounts to about 1 % of the gross local transfer rate $\int |\dot{\sigma} A|\, d\boldsymbol{k}$ for the Superbee scheme, consistent with the $\mathcal{O}(0.5\,\%)$ energy overshoot observed for such a packet. First-order upwind advection is far worse (about 12 %, since its frequency diffusion moves action across the σ gradient), an unlimited second-order flux reduces the defect to about 0.2 %. The x-sweep is exactly energy-neutral for a zonal jet; the defect resides in the y, ν and θ sweeps.

The residual is removed exactly by a conservative energy fixer (attribute `energy_consistent` in `propagation.py`). Writing the discrete advection in flux form per phase-space grid cell i as $\partial_t(A_i \Delta_i) = -\sum_{faces} \delta F_i$, with the phase-space volume Δ_i of the cell and F the total (limited) action transport through a cell face in $\boldsymbol{x}$, ν or θ, the transports cancel pairwise in the sum over all cells and action is conserved to machine precision for any advection scheme. Multiplying by Ω_i, summing over all grid cells and shifting the sums equivalent to a discrete partial integration yields

$$\partial_t \sum_i \Omega_i A_i \Delta_i = \sum_i \sum_{faces} F_i\, \delta\Omega_i \equiv D \tag{35}$$

with the forward difference $\delta\Omega$ of Ω across each cell face. Boundary terms vanish for closed and periodic boundaries (not for reflective ones, for which no correction is made). The

possible leakage D is split into contributions $\sum_{j,k} D_{j,k} = D$ per water column at discrete position (j,k), with the shared horizontal face terms split half-half between neighbouring columns, and each $D_{j,k}$ is cancelled by a small corrective action transport $\tilde{F}_m$ across the frequency faces of that column,

$$\tilde{F}_m = \lambda \, \bar{A}_m \, \delta_\nu \Omega_m \quad , \quad \lambda(\boldsymbol{x}) = -\frac{D_{j,k}}{\sum_m \bar{A}_m \, (\delta_\nu \Omega_m)^2} \tag{36}$$

with m running over all ν faces and directions θ at position (j,k), and the weight $\bar{A}_m$ given by the face average of the action content of the two adjacent cells. Being a transport, $\tilde{F}$ leaves the action conservation untouched, but by the same partial integration it changes the energy by $\sum_m \tilde{F}_m \, \delta_\nu \Omega_m = -D_{j,k}$. The frequency direction is chosen for the corrective flux since it is the only one in which $\delta\Omega$ does not vanish without currents ($\delta_\nu \Omega \approx \delta_\nu \sigma > 0$ for small Doppler shift), and it is where the leakage is generated by action crossing the frequency gradient of Ω. The factor $\bar{A}_m$ ties the correction to the level of action, so that it acts as a small correction of the frequency propagation velocity rather than as a source term and positivity is not threatened, and the factor $\delta_\nu \Omega_m$ makes the denominator sign-definite, so that λ exists whenever action is present at that position. Minimising the action-weighted amplitude $\sum_m \tilde{F}_m^2 / \bar{A}_m$ subject to this constraint yields the same form by the standard Lagrange-multiplier condition: among all conservative frequency-flux corrections that restore the energy, Eq. (36) is the smallest one in the action-weighted least-squares sense, i.e. the correction perturbs the solution as little as possible.

The correction is local (one scalar per position $\boldsymbol{x}$, no global communication) and dormant without currents: for $\boldsymbol{U} = 0$ we have $\Omega = \sigma$, no frequency transports exist, and $D_{j,k} = 0$ so that $\lambda = 0$. In tests the correction amounts to about 1 % of the frequency transports, the tendency-level defect drops from 0.9 % of the gross transfer to machine precision, and in long integrations of a wave packet through a jet the energy $\int \Omega A \, d\boldsymbol{k}$ is conserved exactly (Adams–Bashforth time stepping preserves this, since each stored tendency is corrected individually and Ω is constant while $\boldsymbol{U}$ is frozen between coupling updates), whereas without the fixer the spurious drift amounts to roughly half of the genuine exchange. The transfer from ocean to waves is thus exact on both sides; a structurally energy-conserving advection would achieve the same without an a posteriori correction and remains an option for future work.

The coupling to the ocean

Stokes velocity, wave stress, and wave stress force are calculated in `./pyOM3/wim/interface.py`. The methods there may be called and results passed to the ocean model. The velocity variable of the coupled ocean model is interpreted as the three-dimensional Lagrangian mean velocity $\boldsymbol{u}_L = \boldsymbol{u} + \boldsymbol{u}_S$, with the Eulerian mean velocity $\boldsymbol{u}$ and the Stokes velocity $\boldsymbol{u}_S$, which is the most natural choice for energetic consistency (it also follows the example of interpreting the model's velocity as the isopycnal thickness weighted velocity to account for the effect of meso-scale eddies). Starting from the Craik–Leibovich equations of motion,

the momentum equation for $\boldsymbol{u}_L$ becomes

$$\partial_t \boldsymbol{u}_L - \boldsymbol{u}_L \times \boldsymbol{\nabla} \times (\boldsymbol{u}_L - \boldsymbol{u}_S) + f\boldsymbol{z} \times \boldsymbol{u}_L = -\boldsymbol{\nabla}\pi + \partial_t \boldsymbol{u}_S + (\partial_z \boldsymbol{\tau}, 0) \tag{37}$$

with the three-dimensional operator $\boldsymbol{\nabla} = (\boldsymbol{\nabla}_{\boldsymbol{x}}, \partial_z)$ and a gradient force $\boldsymbol{\nabla}\pi$ collecting the pressure gradient and wave-driven gradient components. The ocean model thus receives the wave-driven force $\partial_t \boldsymbol{u}_S$, the vortex force

$$\boldsymbol{F} = -\boldsymbol{u}_L \times \boldsymbol{\nabla} \times \boldsymbol{u}_S \tag{38}$$

with $\boldsymbol{u}_L \cdot \boldsymbol{F} = 0$, and the wave momentum flux (stress), either as the surface stress $\boldsymbol{\tau}_o$ into the uppermost grid cell or as the interior stress force $\partial_z \boldsymbol{\tau}$; the tracers are advected by $\boldsymbol{u}_L$. Note that the Coriolis term acts on the full $\boldsymbol{u}_L$: with $w = 0$, identical results are obtained using $\boldsymbol{u}$ as model velocity and adding the Stokes–Coriolis force $-f\boldsymbol{z} \times \boldsymbol{u}_S$ instead of $\partial_t \boldsymbol{u}_S$, but with $w \neq 0$ the Lagrangian formulation is required to stay energetically consistent. The scalar product of Eq. (37) with $\boldsymbol{u}_L$ contains no transfer terms related to the vortex force and the Coriolis–Stokes force; the direct transfer to or from the waves is by $\boldsymbol{u}_L \cdot \partial_t \boldsymbol{u}_S$ and by the stress work $\boldsymbol{u}_h^L \cdot \partial_z \boldsymbol{\tau}$, which together form the exchange C of Eq. (3).

Discrete vortex force for horizontally uniform $\boldsymbol{u}_S$ (nh_simple)

For a horizontally uniform Stokes velocity $\boldsymbol{u}_S = (u_S(z), v_S(z), 0)$, as used in the non-hydrostatic model nh_simple, the wave-driven force of Eq. (38) becomes

$$\boldsymbol{F} = -\boldsymbol{u}_L \times \boldsymbol{\nabla} \times \boldsymbol{u}_S = \begin{pmatrix} w\,\partial_z u_S \\ w\,\partial_z v_S \\ -u\,\partial_z u_S - v\,\partial_z v_S \end{pmatrix} \quad , \quad \boldsymbol{u}_L \cdot \boldsymbol{F} = 0 \tag{39}$$

writing $\boldsymbol{u}_L = (u, v, w)$ in this section for brevity. On the C-grid, u, v and w are staggered at the cell faces $(i + \frac{1}{2}, j, k)$, $(i, j + \frac{1}{2}, k)$ and $(i, j, k + \frac{1}{2})$, and the vertical Stokes shear is placed on the w-levels,

$$S^u_{k+\frac{1}{2}} = \frac{u_{S,k+1} - u_{S,k}}{\Delta z} \quad , \quad S^v_{k+\frac{1}{2}} = \frac{v_{S,k+1} - v_{S,k}}{\Delta z} \tag{40}$$

with $S^u = S^v = 0$ at the rigid lid and the flat bottom. Introducing the two-point averages $\overline{\phi}^x_{i+\frac{1}{2}} = (\phi_i + \phi_{i+1})/2$ and analogous in y and z, the discrete force $\boldsymbol{F} = (F^u, F^v, F^w)$ is

$$F^u_{i+\frac{1}{2},j,k} = \overline{\overline{w}^x\, S^u}^z \quad , \quad F^v_{i,j+\frac{1}{2},k} = \overline{\overline{w}^y\, S^v}^z \quad , \quad F^w_{i,j,k+\frac{1}{2}} = -\,\overline{\overline{u}^x}^z S^u - \overline{\overline{v}^y}^z S^v \tag{41}$$

i.e. the products $\overline{w}^x S^u$ are formed on the cell edges $(i+\frac{1}{2}, j, k+\frac{1}{2})$ and averaged back to the u-points, while for F^w the horizontal velocities are averaged first to the cell centers and then to the w-levels where the shear is located. The averaging operators in Eq. (41) are pairwise transposed: each average that moves w towards u in F^u is mirrored by an average moving u towards w in F^w. With the summation-by-parts identity for the two-point average

$$\sum_i \psi_{i+\frac{1}{2}}\, \overline{\phi}^x_{i+\frac{1}{2}} = \sum_i \phi_i\, \overline{\psi}^x_i \tag{42}$$

which holds without boundary terms in the periodic horizontal directions, and in the vertical because the boundary contributions carry the factor $S^u = S^v = 0$ at the lid and bottom faces, the total work of the discrete force vanishes exactly,

$$\sum_{i,j,k} (uF^u + vF^v + wF^w) = 0 \tag{43}$$

for arbitrary $\boldsymbol{u}_L$ and arbitrary profiles $u_S(z)$, $v_S(z)$; this has been verified numerically to machine precision for random fields. Note that the work vanishes only in the sum.

The vanishing work is a property of the spatial operator and does not by itself guarantee numerical stability. Since $\boldsymbol{F}$ is perpendicular to $\boldsymbol{u}_L$, it turns the velocity vector without changing its magnitude, i.e. it acts locally as a rotation with angular frequency $|\partial_z \boldsymbol{u}_S|$, analogous to inertial oscillations with frequency f. An explicit third-order Adams–Bashforth time stepping scheme is stable for such neutral oscillations only if the angle covered per time step remains below $|\partial_z \boldsymbol{u}_S| \Delta t \lesssim 0.72$. Since the Stokes shear is largest in the uppermost grid cell and grows with the developing sea, this number should be monitored during integrations together with the advective Courant numbers.

Discrete vortex force in the hydrostatic case (pe_simple)

In the hydrostatic model pe_simple the Stokes velocity $\boldsymbol{u}_S = (u_S, v_S, 0)$ varies in all three dimensions, and the force Eq. (38) is truncated to

$$\boldsymbol{F} = \begin{pmatrix} -v\,\zeta_S \\ u\,\zeta_S \end{pmatrix} \quad , \quad \zeta_S = \partial_x v_S - \partial_y u_S \quad , \quad \boldsymbol{u}_h^L \cdot \boldsymbol{F} = 0 \tag{44}$$

with $\boldsymbol{u}_h^L = (u, v)$ in this section for brevity. The Stokes-shear terms $w\partial_z u_S$ and $w\partial_z v_S$, part of the first and second component of the total force $-\boldsymbol{u}_L \times \boldsymbol{\nabla} \times \boldsymbol{u}_S$, respectively, are deliberately omitted. This is because the scalar product of these terms with $\boldsymbol{u}_h^L$ is only cancelled by w times the vertical force component $-u\partial_z u_S - v\partial_z v_S$, which the hydrostatic model does not carry. Retained alone, the Stokes-shear terms would thus perform the uncancelled work $w\,(u\partial_z u_S + v\partial_z v_S)$ on $\boldsymbol{u}_L$. The omitted terms are the so-called Stokes shear force of Suzuki and Fox-Kemper (2016). Their neglect is also in agreement with quasi-geostrophic scaling where the Rossby number Ro becomes small: within the Stokes layer of depth h_S the vertical velocity of balanced flow is limited by the surface kinematic condition and its $O(Ro)$ divergence to $|w| \lesssim Ro\,U h_S/L$, such that $|w\partial_z u_S| \sim Ro\,U U_S/L$ independent of h_S and smaller by $O(Ro)$ than the retained vortex force.

On the C-grid, u_S and v_S are given at the cell centers and averaged to the u- and v-points, and the Stokes vorticity is formed at the cell corners $(i+\frac{1}{2}, j+\frac{1}{2})$ by centered differences,

$$\zeta_S = \frac{\overline{v_S}^{\,y}_{i+1} - \overline{v_S}^{\,y}_{i}}{\Delta x} - \frac{\overline{u_S}^{\,x}_{j+1} - \overline{u_S}^{\,x}_{j}}{\Delta y} \tag{45}$$

The products are formed at the corners first, and then averaged back to the velocity points,

$$F^u_{i+\frac{1}{2},j} = -\overline{\zeta_S\,\overline{v}^{\,x}}^{\,y} \quad , \quad F^v_{i,j+\frac{1}{2}} = \overline{\zeta_S\,\overline{u}^{\,y}}^{\,x} \tag{46}$$

with $\boldsymbol{F} = (F^u, F^v)$. Applying the summation-by-parts identity Eq. (42) once in y to the first and once in x to the second term of the work moves both averages onto the velocities, and the two contributions meet at the corners,

$$\sum u F^u + \sum v F^v = \sum_{\text{corners}} \zeta_S \left(-\overline{v}^x \, \overline{u}^y + \overline{u}^y \, \overline{v}^x \right) = 0 \tag{47}$$

i.e. the cancellation is not only global as in nh_simple but holds corner by corner. The boundary terms of Eq. (42) vanish because the velocities vanish on the channel walls and on land, so the identity holds in the presence of topography; this has been verified to machine precision for random fields on a masked channel.

Discrete aspects of the momentum coupling

The discretisation of the remaining coupling terms in the ocean model (C-grid) is chosen such that the energetic properties of the continuous system are preserved where possible, and monitored where not.

Stokes–Coriolis force. Continuously, the combined rotational forces are $(f + \zeta)\hat{z} \times \boldsymbol{u}_L$, which does no work against $\boldsymbol{u}_L$. For the Coriolis part this property carries over exactly to the discrete level if the Stokes velocity is first interpolated from T to the U/V points and then passed through the *same* discrete Coriolis operator C as the Eulerian velocity (the energy-conserving form of the main code, with f evaluated at T points inside the adjoint-paired averaging). By linearity the total discrete Coriolis force is then $C(\boldsymbol{u}_L)$, and the antisymmetry of the operator gives $\sum \boldsymbol{u}_L \cdot C(\boldsymbol{u}_L) = 0$ to machine precision for arbitrary $f(y)$. Note that adding instead a separate Stokes–Coriolis term with its own (single) T-to-U/V averaging leaves an $O(\Delta x^2)$ residual in the $\boldsymbol{u}_L$ energy budget, since $\boldsymbol{u}_S$ then enters the budget once as a T-point field and once through a double interpolation; the difference is a commutator between the 1-2-1 smoothing and f.

Vortex force. With the discretisations of the previous section the $\boldsymbol{u}_L$-work of the applied vortex force vanishes exactly (globally in nh_simple, corner by corner in pe_simple). The coupled driver nevertheless computes $\sum \boldsymbol{u}_L \cdot \boldsymbol{F}$ (volume integrated, with the vortex force exactly as applied) every coupling step and stores it in the coupling output, so that it can be verified to remain at round-off level.

Stokes advection of tracers. The advection of buoyancy by $\boldsymbol{u}_S$ is added in flux form: $\boldsymbol{u}_S$ is averaged from T to the U/V faces, the vertical Stokes velocity w_S is diagnosed from the discrete Stokes continuity equation integrated upward from the bottom ($w_S(-D) = 0$), and the fluxes are differenced like the model's own tracer fluxes, with zero buoyancy flux through the surface face. This form is exactly conservative (the fluxes telescope, so $\int b \, dV$ is preserved to round-off) and constancy-preserving in the interior (a constant b yields zero tendency, since the diagnosed w_S closes the three-dimensional Stokes divergence discretely); both properties were verified to machine precision. In the top cell the surface no-flux condition places the convergence of the Stokes transport (the Stokes pumping w_S at the surface face) into the cell budget, which is the physically correct location: it is compensated by the Eulerian anti-Stokes flow driven by the Stokes–Coriolis force. Centered fluxes are

used; the Stokes Courant number is $O(10^{-3})$, far below the regime where flux limiting matters.

Stress and vertical placement. The wave stress is kinematic (the spectrum is the variance density, so no density factor appears; the air–water density ratio enters through the growth rate of the input source term) and is applied either to the surface face of the top wet cell ($\tau/\Delta z$) or depth-distributed with the same cell-integrated vertical kernel that underlies the Stokes velocity and the Doppler projection, which makes the discrete energy-exchange identity (the C-closure of the previous section) hold to machine precision. Note that with a top-cell thickness large compared to the Stokes decay scale $1/(2k)$, the stored top-cell Stokes velocity is a cell average and correspondingly smaller than the surface value; resolving the near-surface Stokes shear requires Δz of a few metres.

High-frequency tail fluxes to the ocean (tail_flux switch). Above the prognostic cutoff ν_c the spectrum is diagnostic ($E \propto \nu^{-5}$, imposed by the tail parameterisation), and its momentum and energy fluxes to the ocean must be added analytically to the resolved-range dissipation integrals. Two formulations are implemented, selected by the attribute `tail_flux`:

(i) `'wamdi'` (default): the WAMDI dissipation rate continued over the tail, $\int_{\nu_c}^{\infty} \gamma_{out}\, E_{hi}\, \sigma\, d\nu$ with $\gamma_{out} \propto \bar{\sigma}(\sigma/\bar{\sigma})^2(\bar{\alpha}/\alpha_{PM})^2$. The integrand decays like ν^{-2} and converges; note that the corresponding integral with the dissipation of Janssen (1994) diverges strongly because of its $(k/\bar{k})^4$ term (its $(k/\bar{k})^2$ term behaves like the WAMDI form), which is why the WAMDI form is used for this term. However, γ_{out} is the whitecapping rate calibrated on the resolved spectrum and underestimates the dissipation that actually maintains the ν^{-5} equilibrium range — the very reason the tail must be imposed diagnostically — so this form systematically underestimates the tail fluxes (in tests by a substantial factor in the momentum flux).

(ii) `'input'`: in the equilibrium range $S_{in}+S_{nl}+S_{out} = 0$ holds, so the flux to the ocean, $-S_{out}$, equals $S_{in} + S_{nl}$. The input part is the analytic integral of the WAMDI growth rate over the tail: with $a(\theta) = 28u_\star 2\pi\cos(\theta-\theta_w)/g$ (the input parameter evaluated at ν_c), the input being active only above $\nu_1 = 1/a$,

$$I_\tau(\theta) = \frac{\epsilon}{4}(2\pi)^2\nu_c^3\, E_c(\theta) \begin{cases} a\nu_c - \frac{1}{2}, & a\nu_c \geq 1 \\ \frac{1}{2}(a\nu_c)^2, & 0 < a\nu_c < 1 \end{cases} \qquad I_\Phi(\theta) = \frac{\epsilon}{4}(2\pi)\nu_c^2 g\, E_c(\theta) \begin{cases} \frac{a\nu_c}{2} - \frac{1}{3}, & a\nu_c \geq 1 \\ \frac{1}{6}(a\nu_c)^3, & 0 < a\nu_c < 1 \end{cases} \quad (48$$

($\epsilon = \rho_a/\rho_w$, $E_c(\theta) = E(\nu_c, \theta)$; continuous at $a\nu_c = 1$; zero upwind; the integrands again decay like ν^{-2}). The same booking applies to the *in-grid* diagnostic bins (between the per-column prognostic cut-off m_0 and the last grid frequency, where the ν^{-5} shape is imposed): there the flux is evaluated discretely as $\gamma_{in}E$ with the actual imposed spectrum, while the prognostic bins keep the true $\gamma_{out}E$ dissipation — booking the in-grid diagnostic range at γ_{out} would be inconsistent with the equilibrium assertion that the imposed tail itself encodes (and underestimates the flux substantially: the in-grid diagnostic range can carry several times the beyond-grid tail). The nonlinear part uses the conservation of the exact transfer over the full spectrum, $\int_{tail} S_{nl}\, d\Omega = -\int_{resolved} S_{nl}\, d\Omega$: the *prognostic*-range integrals of the stored S_{nl} (momentum weight σ, energy weight g) are credited to

the ocean fluxes (the "cascade credit"; the in-grid diagnostic S_{nl} is part of the equilibrium booking there, and the algebra of the two ranges collapses exactly to the prognostic-masked integral). On the truncated grid the discrete DIA is not exactly conservative at the cutoff; the residual is deliberately included in the credit — whatever the discrete operator removes from the resolved range is deposited in the tail. For the depth-distributed stress (method `calc_stress_work`, as opposed to the surface stress of `calc_stress`) the tail is deposited with its exact profile: with $b = 8\pi^2/g$, $t = -z$, $\beta_L = b\,t\nu_L^2$, the cumulative-from-surface tail stress is $V(t) = C_0[\,a/\nu_L - 1/(2\nu_L^2) - (aJ_2 - J_3)\,]$ with $J_2 = e^{-\beta_L}/\nu_L - \sqrt{\pi bt}\,\mathrm{erfc}\sqrt{\beta_L}$ and $J_3 = \frac{1}{2}(e^{-\beta_L}/\nu_L^2 - bt\,E_1(\beta_L))$, so the cell weights $(V(t_{bot}) - V(t_{top}))/\Delta z$ sum to the surface tail exactly; the cascade credit is deposited in the uppermost cell (its carrier waves have deposition scales $g/2\sigma^2$ of centimetres).

Dissipation energy flux to the ocean. The term $\int \sigma S_{out}\, d\boldsymbol{k}$ of the wave energy budget Eq. (3) is the energy the breaking waves hand to near-surface turbulence. It is diagnosed locally as

$$\Phi_{oc}(\boldsymbol{x}) = -\,g \int \dot{E}_{out}\, d\nu\, d\theta \;\geq\; 0 \tag{49}$$

(kinematic; $\rho_w\Phi_{oc}$ in $\mathrm{W\,m^{-2}}$; WAM's PHIOC), computed from the same stored dissipation rate γ_{out} and with the same spectral measures as the wave stress, and written to the coupling output both as the local field and as its domain integral in the energy-diagnostic record. The dissipation of the diagnostic ν^{-5} tail is included according to the `tail_flux` switch described above (under `'input'` together with the nonlinear cascade credit). The field is the surface boundary condition for the flux F of the turbulent kinetic energy (TKE) closure following Gaspar et al. (1990) in the ocean component ($\rho_w\Phi_{oc}$ as the surface TKE flux, in place of the $u_\star^3$-based parameterisation), which routes this term into the ocean's turbulent kinetic energy budget and thereby closes the coupled energy cycle. The TKE closure receives, besides the usual shear production by the Eulerian mean flow, a shear production term related to the mean Stokes velocity, $c_u K(\partial_z \bar{\boldsymbol{u}}) \cdot \partial_z \bar{\boldsymbol{u}}_S$, resulting from the down-gradient closure of the eddy stresses; the corresponding energy is exchanged with the mean Lagrangian kinetic energy.

Vertical friction and mixing. For consistency with the TKE closure, the vertical friction operator acts on the Eulerian mean velocity, $\partial_z A_v \partial_z(\boldsymbol{u}_L - \boldsymbol{u}_S) = \partial_z A_v \partial_z \boldsymbol{u}$: the work of this viscous term against $\boldsymbol{u}_L$ then exactly matches the shear production terms of the TKE closure, including the Stokes-shear production. Mixing the Lagrangian instead of the Eulerian momentum would break this energy consistency.

Coupling frequency and the exchange term. The two models exchange fields once per coupling interval Δt_c, chosen as a common multiple of the two time steps: the component with the smaller time step sub-cycles $\Delta t_c/\Delta t$ steps between exchanges. When the wave step is the smaller one, $\Delta t_c = \Delta t_{\mathrm{ocean}}$ and the wave model sub-cycles; when $\Delta t_{\mathrm{ocean}} < \Delta t_{\mathrm{wave}}$ (e.g. a non-hydrostatic or very high resolution ocean) the roles reverse and the ocean sub-cycles, with $\Delta t_c = \Delta t_{\mathrm{wave}}$. In either case all coupling fields are piecewise constant over the coupling interval, so the exchange work $C = \int \boldsymbol{u}_L \cdot \partial_t \boldsymbol{u}_S\, dV$ is evaluated with $\partial_t \boldsymbol{u}_S$ as the difference quotient across the coupling update, i.e. over Δt_c rather than over the time

step of either component. The domain-integrated energy diagnostics — wind input to the waves $g \int \gamma_{\rm in} E \, d\Omega \, dA$, the stress work on the ocean (computed with the identical stencils as the applied forcing), the exchange term C, and the vortex-force work — are written to the coupling output at every coupling step, so the discrete energy pathway of the coupled system is monitored throughout the integration rather than assumed. The choice of Δt_c is constrained by the physics rather than by stability: the coupling fields should be well sampled, i.e. Δt_c small compared to the evolution time scales of the exchanged quantities (translation of the wind systems across a grid cell for the stress and Stokes fields, and the evolution of the resolved ocean currents for the refraction terms).

Propagation, boundary conditions, land mask

Action is propagated since it avoids to calculate the energy exchange with the mean flow explicitely. Advection uses the second-order scheme with Superbee limiter (`tracer_flux_superbee`) taken from MITgcm. A first-order upwind option is also available (`tracer_flux_upwind`, selected by the attribute "f_limiter"="upwind"); it shares the identical finite-volume (face-flux) structure — the same land masking, halo exchange, closed frequency faces and reflection — so conservation and the energy fixer of Eq. (36) still apply, but the flux limiter is set to zero. Upwind is more diffusive (it smears sharp swell fronts) but robust to grid-scale noise. The refraction drift velocities $\dot{\nu}$ and $\dot{\theta}$ can optionally be clipped to a CFL bound (attribute `limit_fdot`, default off, with the bound set by `refraction_cfl`, default 0.5): strong current shear can otherwise drive the spectral advection beyond its stability limit locally. The clipping breaks the exact ray equations where it engages and is meant as a safeguard, not a substitute for a proper time step.

Lateral boundary conditions in x and y are periodic (attribute "periodic_in_x"/"periodic_in_y"), closed, or reflective. Boundary conditions in frequency are always closed, in direction always periodic. Lateral reflection boundary conditions at coast lines are implemented the same way as in the IDEMIX2 in pyOM2. These are activated by setting the attribute "enable_reflection" to true value'. If undefined or false, the default of closed boundary are used where energy may pile up near the boundary. Land mask is set in the template method "set_topography" in `./pyOM3/wim/main.py`.

Time stepping scheme

Each time step is split into an explicit propagation substep for the wave action A followed by an implicit source substep for the wave energy $E = \sigma A$. The propagation tendency $\dot{A}|_{\rm prop}$ (the sum of the flux divergences in x, y, ν and θ) is advanced with a multi-step Adams–Bashforth scheme,

$$A^{n+1} = A^n + \Delta t \left(c_A \, \dot{A}^{\,n}|_{\rm prop} + c_B \, \dot{A}^{\,n-1}|_{\rm prop} + c_C \, \dot{A}^{\,n-2}|_{\rm prop} \right). \tag{50}$$

The scheme used is the stabilised second- or third-order Adams–Bashforth pair

$$(c_A, c_B, c_C) \;=\; \left(\frac{3}{2} + \epsilon, \; -\frac{1}{2} - \epsilon, \; 0 \right), \qquad \epsilon \approx 0.05, \quad \text{for second order} \tag{51}$$

$$(c_A, c_B, c_C) \;=\; \left(\frac{23}{12},\; -\frac{16}{12},\; \frac{5}{12}\right) \quad \text{for third order} \tag{52}$$

both started with a forward Euler step $(1, 0, 0)$. The time loop uses the AB2+ϵ pair; the third-order coefficients are given for reference but are deliberately not used (see below). The recent tendencies are held in a rotating three-level buffer (the pointers "tau", "taum1", "taum2"). Only the propagation is integrated this way; the source terms are treated separately below. After the propagation update the action is subject to the lateral boundary conditions and, on land, is masked to zero.

The choice of the time scheme for the propagation is more delicate than usual because the advection uses a flux *limiter*. TVD limiters (Superbee, upwind) guarantee a monotone, non-oscillatory update — in particular $A \geq 0$ — only when the limited flux is applied in a *single forward Euler step* (or a strong-stability-preserving Runge–Kutta combination of such steps). Multi-step Adams–Bashforth schemes combine tendencies from several time levels with *negative* weights, which destroys this guarantee at any CFL number: each stored tendency is limited with respect to its own state, but the negatively-weighted combination is free to over- and undershoot. For the third-order scheme (23/12, −16/12, 5/12) the negative weight is large and the violation is severe: a smooth, sign-definite action field develops undershoots ($A < 0$) in an alternating grid-scale ($2\Delta x$) checkerboard pattern at sharp fronts (storm edges, swell fronts). Unlike in the ocean model — where the same combination is harmless because tracers are not sign-definite, explicit (biharmonic and vertical) diffusion damps grid-scale noise every step, and no term amplifies small wiggles — the wave model has no explicit diffusion, its action must remain non-negative, and the source terms amplify any ± pattern strongly non-linearly (through $\alpha = \bar{E}\,\bar{\sigma}^4$ and the cubic DIA transfer), so the checkerboard grows over $\mathcal{O}(1)$ day until floating-point overflow. The AB2+ϵ coefficients Eq. (51) reduce the negative weight to $-(1/2 + \epsilon)$, which in practice (CFL $\lesssim 0.25$) is small enough for the limiter's built-in slack to absorb the violation: the action remains non-negative to machine precision in long integrations where AB3 fails within a day. This is an empirical rather than a rigorous property; forward Euler $(1, 0, 0)$ or an SSP Runge–Kutta scheme restores the formal TVD guarantee at the cost of temporal accuracy or an additional flux evaluation per step.

Two safeguards back this up. First, a positivity floor (attribute `floor_action`, default on) clips the action at zero after each propagation step, as is standard in WAM. The clipped mass — the action the floor removes, which equals the spurious action it injects and hence the energy-consistency cost of the clip — is measured *before* clipping and a warning is issued whenever it exceeds round-off scale relative to the total action, so a silently active floor cannot mask a genuine positivity violation; switching the floor off restores the raw scheme. Second, $\min A$ is monitored automatically every $O(100)$ steps against a relative round-off threshold and a warning names the remedy (larger ϵ or forward Euler): any limiter/time-scheme violation shows up there long before a blow-up.

Writing the source terms as growth rates acting on the energy, the source substep integrates $\partial_t E = S$ with $S = \dot{E}_{nl} + (\gamma_{in} + \gamma_{out})E$, where $\dot{E}_{nl}$ is the non-linear (DIA) transfer. A fully explicit update would require a very small time step because the growth

rate increases with frequency (this is also the reason for the high-frequency cut-off). Instead a linearised-implicit (semi-implicit) step is used: linearising S about the current state, $S(E^{n+1}) \approx S(E^n) + \partial_E S\,(E^{n+1} - E^n)$, and solving for E^{n+1} gives

$$E^{n+1} = E^n + \frac{S\,\Delta t}{1 - \Delta t\,\partial_E S}, \qquad \partial_E S = \gamma_{in} + \gamma_{out} + \partial_E \dot{E}_{nl}, \tag{53}$$

where the diagonal of the non-linear functional derivative $\partial_E \dot{E}_{nl}$ is returned by the DIA backend alongside the transfer itself. This is unconditionally stable for a purely dissipative feedback ($\partial_E S < 0$), for which the denominator exceeds unity and the update is damped. The growth can additionally be bounded as in WAM by a growth limiter (attribute `limit_growth`, default off) that caps S (not $\partial_E S$) so that the energy cannot grow by more than a prescribed fraction per step,

$$S \leftarrow \min\!\big(S,\ 10^{-7}\, g\, \max(u_\star,\, u_\star^{\min})\, \bar{\sigma} \,/\, (2\pi\nu^4)\big)\,, \qquad u_\star^{\min} = 0.01\ \mathrm{m\,s^{-1}}. \tag{54}$$

The friction-velocity floor $u_\star^{\min}$ in the cap is essential and easy to overlook: without it, the cap vanishes wherever the wind vanishes and the limiter degenerates to $S \leftarrow \min(S, 0)$ in calm cells. This clips only the *positive* lobes of the nonlinear transfer, destroying the conservative $\pm$ structure of S_{nl} exactly there, and acts as a small but systematic negative-action source. In a calm region that accumulates swell without a sink (e.g. a walled basin with reflective coasts outside the wind band), the resulting violation grows linearly in time — independent of the time step and of the time-stepping scheme — until, after $O(100)$ days, the nonlinear source terms of the accumulated spectrum activate and the integration fails abruptly. The floor keeps the cap strictly positive everywhere while leaving the physics unchanged: $u_\star^{\min}$ is far below the input threshold of the slowest resolved wave ($c \approx 28\,u_\star$ requires $u_\star \gtrsim 0.06\ \mathrm{m\,s^{-1}}$ for the highest resolved frequency), so the wind input remains identically zero in floored cells; only the growth cap becomes a finite positive bound instead of zero. Because the limiter bounds only the numerator, the denominator $1 - \Delta t\,\partial_E S$ can still vanish or change sign where $\partial_E S$ is large and positive — most notably at high frequency during spin-up from a small seed, where the DIA derivative is steep. When this happens the update overshoots and the highest bins become unstable. Two safeguards are used: the high-frequency cut-off removes the steepest bins from the prognostic integration (replacing them by the diagnostic ν^{-5} tail), and, optionally, only the dissipative part of the feedback is kept in the implicit denominator,

$$\partial_E S \ \rightarrow\ \min(\partial_E S,\, 0)\,, \tag{55}$$

guaranteeing $1 - \Delta t\,\partial_E S \geq 1$. This is activated by setting the attribute `implicit_dissipation_only` to `True` and equals the standard WAM/WW3 construction — the stabilising dissipative feedback is treated implicitly while the growth is treated explicitly and bounded by the limiter. It leaves the equilibrium ($S = 0$) unchanged and therefore does not affect the energy balance; it only alters the transient integration of the growth phase.

After the source substep the energy is masked to zero on land, the high-frequency tail is re-imposed through `imphftail`, and the action is recovered as $A = E/\sigma$ for the next propagation step.

The time step of the propagation is limited by the advective CFL number based on the fastest group velocity, i.e. the lowest resolved frequency (Superbee is stable to CFL $\lesssim 0.5$, upwind to CFL < 1; in practice CFL $\approx$ 0.2–0.25 is used, which is also the regime in which the AB2+ϵ scheme preserves the non-negativity of the action, see the time-stepping section). Note that the grid-scale checkerboard described there is *not* a CFL violation — it appears at any Courant number when a multi-step scheme with a large negative weight (AB3) is combined with the flux limiter — so it cannot be cured by reducing the time step; the cure is the time scheme. Substantially larger time steps than the advective CFL allows would require an implicit or semi-Lagrangian treatment of propagation, at the cost — for plain semi-Lagrangian — of the exact (face-flux) energy conservation.

Model parameters and options

All parameters and options are attributes of the model instance. They are set either through the parameter dictionary passed at construction or by assigning the attribute on the instance before the model is stepped; options marked with a default may simply be left unset. The tables below list every attribute the wave model reads, with its default where one exists.

Required parameters (no default)

Attribute	Unit	Meaning
`nx`, `ny`	—	global horizontal grid size (cells, without halo points)
`dx`, `dy`	m	horizontal grid spacing
`nf`	—	number of frequency bins (local on every PE)
`nd`	—	number of direction bins (local on every PE)
`f0`	1/s	lowest frequency of the geometric frequency grid (WAM value 0.035)
`df0`	—	relative frequency increment $\Delta\nu/\nu$ of the geometric grid (WAM value 0.1)
`dt`	s	time step
`periodic_in_x`	—	lateral boundary in x: periodic if true, closed if false (the outermost ring of cells is then set to land)
`periodic_in_y`	—	as above for y

The constructor additionally takes the keyword arguments `n_pes_x`, `n_pes_y` (processor grid, default 1×1; their product must equal the communicator size), `halo_points` (default 1) and `comm_in` (default: the MPI world communicator).

Forcing and coupling fields

Attribute	Default	Meaning
`ustar`	required	friction velocity magnitude $u_\star$; scalar or horizontal field (scalars are broadcast)
`udir`	required	wind direction θ^* (radians); scalar or field
`ocean_U`, `ocean_V`	unset	Doppler velocity $\boldsymbol{U}(\nu)$ provided by the coupler. If unset, all current effects (advection by currents, refraction, the energy fixer) are dormant
`ocean_Udf`, `ocean_Vdf`	0	frequency derivative $\partial_\nu \boldsymbol{U}$ of the Doppler velocity (radial group-velocity correction R and frequency refraction)

Options with defaults

Attribute	Default	Meaning
`snonlin_backend`	`'fortran'`	DIA implementation: `'fortran'` (original WAM code; requires double precision), `'jax'` or `'numpy'`; all give identical results
`grav`	9.81	gravitational acceleration
`rho_o`	1024	reference water density. Currently unused inside the wave model: the air–water density ratio of the source terms is fixed at 1.225/1000
`z0`	0.0002	roughness length; read only by the optional WAM input parameterisation, inactive with the default WAMDI input
`mask_energy`	off	prognostic cut-off ν_c per column with the imposed ν^{-5} diagnostic tail (`imphftail`); prerequisite for the equilibrium booking of `tail_flux = 'input'`
`limit_growth`	off	WAM growth limiter, Eq. (54)
`implicit_dissipation_only`	off	keep only the dissipative part of the source feedback in the implicit denominator, Eq. (55)
`floor_action`	on	positivity floor $A \leftarrow \max(A, 0)$ after the propagation step, with the clipped mass monitored and warned about
`f_limiter`	`'superbee'`	advection flux limiter: `'superbee'` (second order) or `'upwind'` (first order, diffusive, robust)
`energy_consistent`	off	conservative energy fixer of the propagation operator, Eq. (36)

Attribute	Default	Meaning
`enable_reflection`	off	specular reflection of the advective fluxes at coastlines
`limit_fdot`	off	clip the refraction velocities $\dot{\nu}, \dot{\theta}$ to a CFL bound
`refraction_cfl`	0.5	the CFL bound used by `limit_fdot`
`closed_bottom`	off	extend the bottom cell of the coupling kernel to $z \to -\infty$ (exact barotropic projection, no momentum lost below the column floor)
`tail_flux`	`'wamdi'`	booking of the diagnostic-tail fluxes to the ocean: `'wamdi'` (dissipation-rate continuation) or `'input'` (equilibrium-range γ_{in} integral plus nonlinear cascade credit)

Snapshot diagnostics

The netCDF snapshot class reads the attribute `snapint` (snapshot interval in seconds, required) and takes the setup arguments `snap_file_name` (default `'snap.cdf'`) and `show_each_time_step` (default on). The land mask is defined by overloading the template method `set_topography`; the default is an all-ocean domain, with closed outer boundaries generated automatically from `periodic_in_x`/`periodic_in_y`.